

SkillVault Project Technical Report

AI-Powered Portfolio and Roadmap Builder

Project Team

Based on Review Paper "SkillVault – AI-Powered Portfolio and Roadmap Builder: A Comprehensive Review"

November 13, 2025

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Chapter 1

Introduction

SkillVault is a pioneering web application designed to bridge the critical gap between theoretical skill acquisition and practical proof-of-work required by modern employers. It is envisioned as a unified platform that integrates personalized career planning, skill tracking, and automatic portfolio generation, fundamentally transforming how individuals approach career development in a rapidly evolving digital economy.

1.1 Objective of the Project

The primary objective of the SkillVault project is to solve the endemic *experience gap* faced by new learners and career changers. This gap exists because existing tools are disconnected: learning platforms (like Coursera or edX) focus on content, while hosting platforms (like GitHub or Behance) focus on display.

- **Unify the Career Journey:** To create a single, cohesive platform where learning, building, and showcasing one's professional worth happen seamlessly.
- **Provide Personalized Guidance:** To utilize Artificial Intelligence (AI) to transform vague career aspirations into clear, step-by-step, time-bound, and personalized learning roadmaps.
- **Mandate Proof-of-Work:** To ensure that skill learning is inextricably linked to project creation, automatically translating completed work into a professional portfolio that serves as *objective proof* of capability.

1.2 Description of the Project

SkillVault functions as an intelligent career coach and portfolio manager. A user initiates their journey by specifying a target job role (e.g., "Data Analyst"). The core system, powered by an AI recommendation engine, then undertakes a comprehensive analysis:

1. **Decomposition:** The target role is broken down into constituent technical, domain, and soft skills.

2. **Sequencing:** Skills are organized into a logical, sequential roadmap grouped into manageable 3-6 week **Milestones**.
3. **Integration:** Each milestone includes specific learning resources, recommended practical projects, and relevant certifications.
4. **Tracking:** As the user marks milestones and projects as complete, the system automatically tracks progress, sends motivational reminders, and updates a public-facing portfolio.

This design directly addresses learner confusion and lack of motivation while simultaneously providing recruiters with verifiable, project-based evidence of a candidate's competence and dedication.

1.3 Scope of the Project

The scope of SkillVault is comprehensive, encompassing the entire journey from beginner skill identification to job readiness and talent discovery.

1.3.1 Inclusions

- **AI-Powered Roadmap Generation:** Initial and dynamic adjustment of learning paths based on user input, existing skills, and market data (e.g., O*NET, job trends).
- **Personalized Learning Tracking:** Granular tracking of skill mastery and overall career progress percentage, coupled with automated coaching and accountability features (reminders, weekly reports).
- **Automatic Portfolio Builder:** Automated aggregation and formatting of completed projects into a polished, professional portfolio accessible via a shareable link.
- **Recruiter Dashboard:** A specialized interface for companies to search candidates based on proven skills, project completion metrics, and long-term dedication, moving beyond traditional resume screening.
- **Core Technology Stack:** Implementation using a modern, scalable architecture: React (Frontend), Python/AI (Backend), and SQL (Database).

1.3.2 Exclusions (Out of Scope for Initial Release)

- **In-House Learning Content Creation:** SkillVault will not create its own course videos or tutorials. It will aggregate and link to external, reputable resources (e.g., Coursera, YouTube, specific books).
- **Financial Transaction Processing:** Direct integration with payment gateways for course enrollment is out of scope. Users will be directed to external resource websites for purchases.

- **Direct Job Application Handling:** While the Recruiter Dashboard enables candidate discovery, SkillVault will not feature a full job board or ATS (Applicant Tracking System) functionality in the initial phase.

1.3.3 Use Case Model

The SkillVault system involves three primary actors: the **Learner**, the **Recruiter**, and the internal **AI System**.

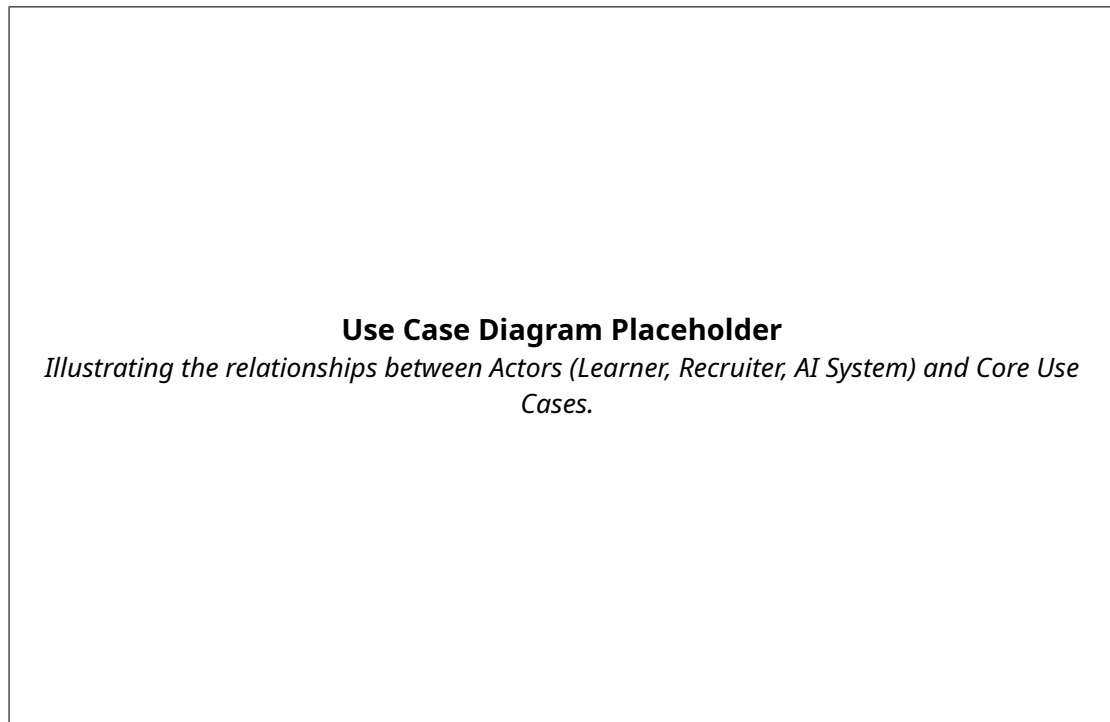


Figure 1.1: High-Level Use Case Model for SkillVault.

Table 1.1: Key Use Cases and Actors

Use Case	Primary Actor	Description
Define Career Goal	Learner	Learner inputs target role (e.g., "Software Developer") and existing skills/schedule constraints.
Generate Personalized Roadmap	AI System	AI analyzes input and market data (O*NET) to structure skills into sequential Milestones with resources and projects.
Track Milestone Progress	Learner	Learner marks a skill/project as "Completed," triggering portfolio updates and progress calculations.

Table 1.1: Key Use Cases and Actors (Continued)

Use Case	Primary Actor	Description
View Learning Resources	Learner	Learner accesses curated links (courses, articles) suggested by the AI for the current milestone.
Publish Professional Portfolio	Learner	Learner configures public visibility and shareable link for their auto-generated project portfolio.
Search and Filter Candidates	Recruiter	Recruiter uses the dashboard to find Learners based on completed projects, validated skills, and progress metrics.
Receive Progress Reports	AI System	AI sends weekly or daily email/in-app reports to the Learner to maintain accountability.

Chapter 2

System Description

The SkillVault system description details the intended users, necessary constraints, and the formalized requirements that govern its functionality and quality.

2.1 Customer/User Profiles

The platform targets two distinct, yet mutually beneficial, user groups.

2.1.1 Profile A: The Learner / Career Changer

- **Goal:** To acquire the necessary skills and, crucially, the *proof* (portfolio) to successfully secure a specific job role (e.g., first job, career transition).
- **Pain Points:** Feeling overwhelmed ("I don't know where to start"), lack of motivation, difficulty connecting theoretical learning to practical projects, and having an empty or weak resume.
- **Expectations:** A clear, step-by-step path; automated tracking; suggestions for specific, valuable projects; and a professional portfolio that builds itself with minimal effort.
- **Key Features Needed:** AI Roadmap, Progress Tracking Dashboard, Project Integration.

2.1.2 Profile B: The Recruiter / Hiring Manager

- **Goal:** To efficiently identify and recruit candidates who have objectively demonstrated job-specific skills through completed projects.
- **Pain Points:** Resumes that all "look the same," difficulty verifying claimed skills, and a long, costly hiring process due to the need for extensive technical interviews.
- **Expectations:** A simple search interface; access to candidates' actual projects (code, design files); visibility into a candidate's learning dedication (progress over time).
- **Key Features Needed:** Recruiter Dashboard, Skill-Based Candidate Search, View Candidate Progress History.

2.2 Assumptions and Dependencies

2.2.1 Assumptions

- **Data Availability:** It is assumed that publicly available, comprehensive job market data (e.g., O*NET, LinkedIn trends) will remain accessible for the AI engine's rule-based system.
- **User Commitment:** The system assumes that the Learner will actively engage with the recommended resources and complete the specified projects for the tracking mechanism to be effective.
- **External Resource Quality:** The quality and availability of external learning resources (Coursera, Udemy, etc.) linked by the AI are assumed to be maintained by their respective providers.

2.2.2 Dependencies

- **Technology Stack Dependencies:** The project is dependent on the stability and maturity of the chosen core technologies: React (Frontend), Python (Backend/AI), and a robust SQL Database.
- **Internet Connectivity:** Both Learners and Recruiters require stable, high-speed internet access to use the web application effectively.
- **Git/External Hosting Integration:** The automatic portfolio builder is dependent on external platforms (like GitHub) for hosting the actual project files (e.g., code repositories).

2.3 Functional Requirements

Functional requirements detail what the SkillVault system *must do* for the users.

Table 2.1: SkillVault Functional Requirements

ID	Requirement Description
FR-1.0	User Authentication: The system shall allow users to register, log in, and maintain a unique user profile securely.
FR-2.0	Goal Definition: The system shall allow a Learner to input a specific career goal (Job Title) and estimate their weekly available study hours.
FR-3.0	Skill Assessment: The system shall allow a Learner to optionally specify existing skills they already possess.
FR-4.0	AI Roadmap Generation: The AI Backend shall generate a personalized, sequenced learning roadmap, categorized into Milestones, based on FR-2.0 and FR-3.0 data.

Table 2.1: SkillVault Functional Requirements (Continued)

ID	Requirement Description
FR-5.0	Resource Recommendation: For each Skill/Milestone, the system shall provide curated links to external learning resources (courses, books, tutorials).
FR-6.0	Project Assignment: Each Milestone shall include at least one recommended practical project designed to validate the learned skills.
FR-7.0	Progress Tracking: The Learner shall be able to manually mark skills and projects as 'Not Started,' 'In Progress,' or 'Completed.'
FR-8.0	Progress Visualization: The system shall display the Learner's overall completion percentage for their career goal via visual dashboards and graphs.
FR-9.0	Automated Portfolio Creation: Upon a project being marked 'Completed,' the system shall prompt the Learner for a description, tools used, and links (e.g., GitHub URL).
FR-10.0	Portfolio Publishing: The system shall automatically aggregate completed projects into a nicely formatted, public-facing portfolio page with a unique shareable URL.
FR-11.0	Portfolio Privacy Control: The Learner shall have granular control over portfolio visibility (Private, Shareable Link Only, Public).
FR-12.0	Recruiter Dashboard Access: The system shall provide a secure, separate interface for registered Recruiters.
FR-13.0	Candidate Search: The Recruiter Dashboard shall allow filtering and searching for candidates based on specific completed skills and project types.
FR-14.0	Progress History View: Recruiters shall be able to view a candidate's historical progress data (e.g., time spent per milestone, consistency).
FR-15.0	Notification System: The system shall send periodic reminders and weekly progress summaries to the Learner via email and in-app notifications.

2.4 Non-Functional Requirements

Non-functional requirements describe the quality and constraints under which the system must operate.

2.4.1 Performance Requirements (NFR-P)

- **NFR-P-1 (AI Response Time):** The initial AI Roadmap Generation (FR-4.0) must complete within a maximum of 5 seconds from the time the user submits their goal.
- **NFR-P-2 (Page Load Time):** All primary user interfaces (Dashboard, Portfolio View) must load fully within 2 seconds on a modern browser using a standard broadband connection.
- **NFR-P-3 (Concurrency):** The system must be able to support 10,000 concurrent active Learners without performance degradation.

2.4.2 Security Requirements (NFR-S)

- **NFR-S-1 (Data Encryption):** All sensitive user data (passwords, emails) must be stored using industry-standard hashing and encryption protocols.
- **NFR-S-2 (Access Control):** Access to private portfolios and the Recruiter Dashboard must be strictly managed through role-based access control.
- **NFR-S-3 (Injection Prevention):** All data inputs, especially those linking to external project URLs, must be rigorously sanitized to prevent Cross-Site Scripting (XSS) and SQL Injection attacks.

2.4.3 Usability Requirements (NFR-U)

- **NFR-U-1 (Responsiveness):** The entire application, built using React, must be fully responsive and functional across desktop, tablet, and mobile devices.
- **NFR-U-2 (Clarity):** The Progress Visualization (FR-8.0) must utilize clear, intuitive charts and graphs that are easily understandable by non-technical users.
- **NFR-U-3 (Consistency):** All navigational elements and component styling must remain consistent across the application interface.

2.4.4 Reliability Requirements (NFR-R)

- **NFR-R-1 (Availability):** The system must maintain an uptime of 99.9% outside of scheduled maintenance windows.
- **NFR-R-2 (Data Integrity):** The SQL Database must utilize transactional integrity to ensure that data updates (e.g., marking a project complete) are accurate and non-corruptible.

- **NFR-R-3 (Error Logging):** The system must maintain a comprehensive log of all critical system errors (database failures, AI engine exceptions) for rapid debugging.

Chapter 3

Design

The SkillVault system design adheres to a scalable, modular architecture to separate concerns and facilitate future development of the AI engine.

3.1 System Design

SkillVault utilizes a standard **Three-Tier Architecture** for web applications:

1. **Presentation Tier (Frontend):** Built with **React**. This is the user interface layer responsible for display, user input handling, and responsiveness. It communicates with the Application Tier via RESTful API calls.
2. **Application Tier (Backend):** Built with **Python** (leveraging its strength for AI/ML). This tier contains the core business logic, including the AI Recommendation Engine, progress calculation logic, and data validation.
3. **Data Tier (Database):** A robust **SQL Database**. This tier is the central memory, storing user profiles, goals, roadmaps, project metadata, and progress logs.

This separation ensures that the AI logic can be updated and scaled independently of the user interface, providing a highly maintainable and high-performance system.

3.1.1 E-R Diagram

The Entity-Relationship (E-R) diagram conceptually models the data structure and relationships within the SkillVault database.

Key Entities and Relationships:

- **User** → (1:N) → **Goal**: One user can define multiple career goals.
- **Goal** → (1:N) → **Milestone**: Each goal is composed of many milestones.
- **Milestone** → (1:N) → **Skill**: A milestone covers multiple required skills.
- **Milestone** → (1:N) → **Project**: A milestone is validated by one or more recommended projects.

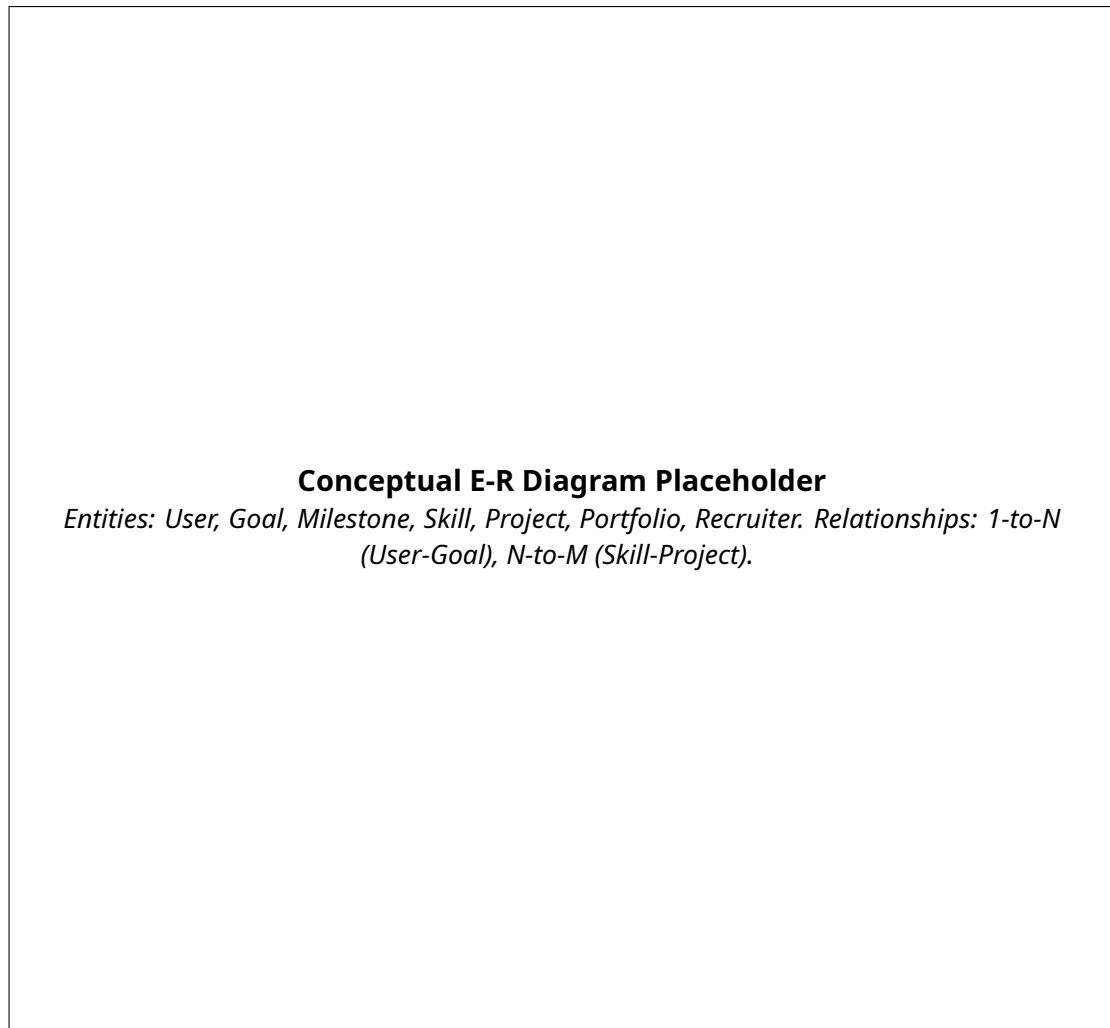


Figure 3.1: Conceptual Entity-Relationship Diagram for SkillVault.

- **Project** → (1:1) → **Portfolio**: A completed project is added as an entry in the user's portfolio.
- **User** → (1:1) → **Recruiter Access**: Recruiters are a special class of User with read-only access to specific public portfolios.

3.1.2 DFD's

Data Flow Diagrams (DFDs) illustrate the movement of data through the system and the processes that transform it.

Data Flow Diagram Level 0 (Context Diagram)

The Level 0 DFD shows the entire SkillVault system as a single process, illustrating its primary inputs and outputs with external entities.

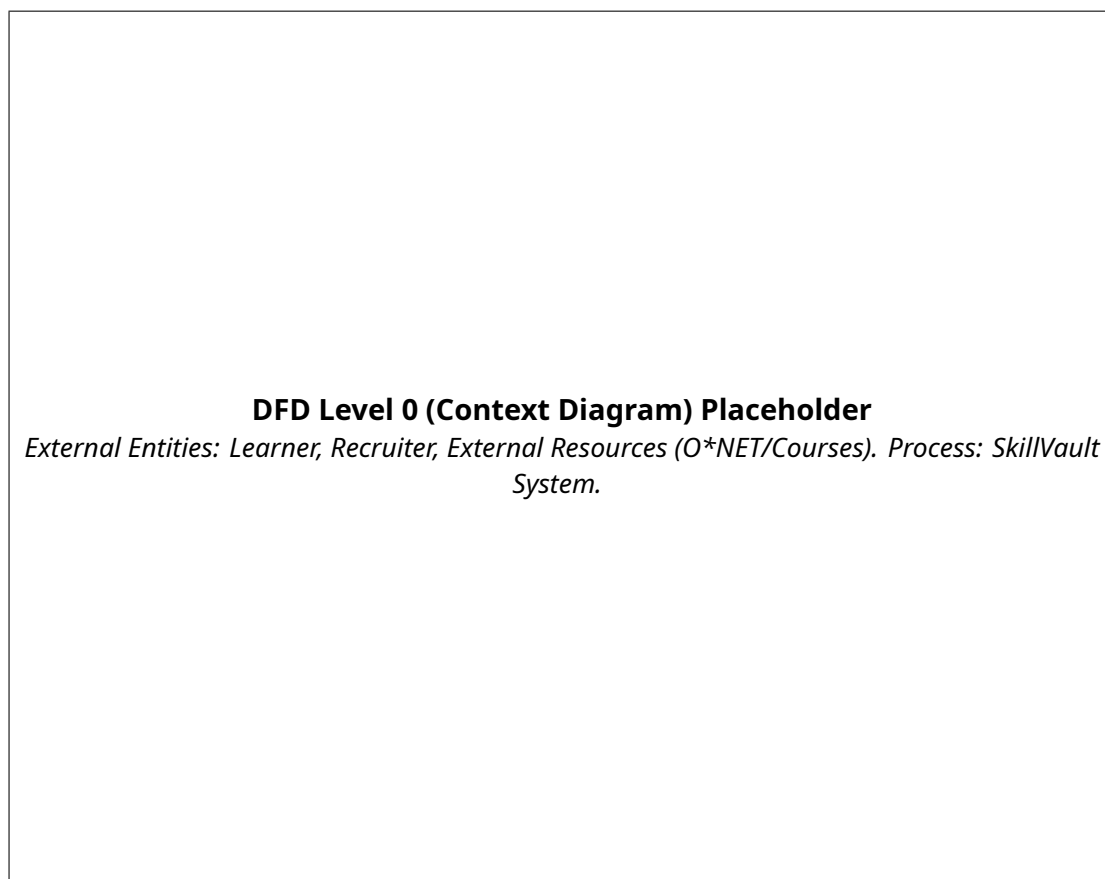


Figure 3.2: Data Flow Diagram Level 0: SkillVault System Context.

- **Input from Learner:** Goal Title, Existing Skills, Completion Status.
- **Output to Learner:** Roadmap, Resources, Portfolio View, Progress Reports.
- **Input from Recruiter:** Search Criteria.
- **Output to Recruiter:** Candidate Profiles, Progress History.

- **External Data Flow:** Job Market Data (In), Resource Links (Out).

Data Flow Diagram Level 1 (Roadmap Generation)

This Level 1 DFD details the core process of generating the personalized roadmap (Process 1.0 from Level 0).

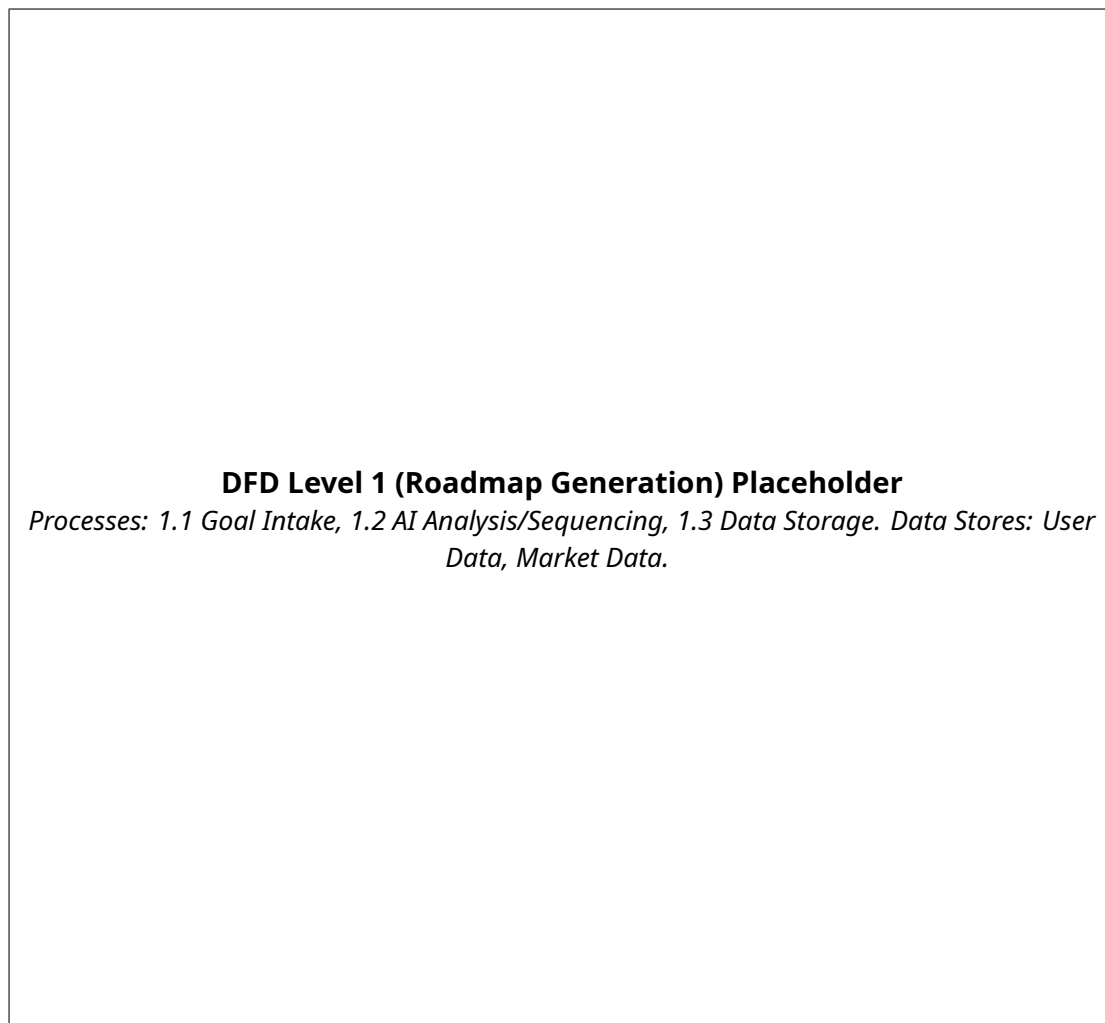


Figure 3.3: Data Flow Diagram Level 1: Personalized Roadmap Generation.

1. **Process 1.1: Goal Intake:** Learner inputs Goal/Skills (Data In). Stores initial goal in **D1: User Data**.
2. **Process 1.2: AI Analysis & Sequencing:** Reads Goal from D1. Queries **D2: Market Data** (O*NET). AI Engine (Python Backend) executes rule-based or ML logic to generate sequenced Milestones, Projects, and Resource Links.
3. **Process 1.3: Data Storage & Delivery:** Writes the complete Roadmap structure to **D3: Roadmap Data**. Delivers the Roadmap to the Learner (Data Out).

3.2 Database Design

The database design, based on the E-R diagram, prioritizes structural integrity (SQL) and speed. Key tables are detailed below.

3.2.1 Users Table

Table 3.1: Schema: Users

Field Name	Data Type	Description
user_id (PK)	INT / UUID	Unique identifier for the user.
email	VARCHAR(255)	User's email (unique, used for login).
password_hash	VARCHAR(255)	Hashed password for secure authentication.
role	ENUM	'Learner', 'Recruiter', 'Admin'.
registration_date	DATETIME	Date and time of account creation.
last_login	DATETIME	Last time the user accessed the system.

3.2.2 Goals Table

Table 3.2: Schema: Goals

Field Name	Data Type	Description
goal_id (PK)	INT	Unique goal identifier.
user_id (FK)	INT / UUID	Foreign key linking to the Users table.
target_job_title	VARCHAR(100)	E.g., 'Software Developer', 'Digital Marketer'.
start_date	DATE	Date the goal was created.
target_completion_date	DATE	Estimated completion date based on AI's estimate.
status	ENUM	'Active', 'Completed', 'Abandoned'.

3.2.3 Milestones Table

Table 3.3: Schema: Milestones

Field Name	Data Type	Description
milestone_id (PK)	INT	Unique milestone identifier.
goal_id (FK)	INT	Foreign key linking to the Goals table.
title	VARCHAR(150)	E.g., 'Python Basics', 'Data Wrangling'.
sequence_order	INT	The order in which the milestone must be completed.
estimated_time_hours	INT	AI's time estimate for completion.
completion_date	DATETIME / NULL	Date completed by the user.

3.2.4 Projects Table

Table 3.4: Schema: Projects

Field Name	Data Type	Description
project_id (PK)	INT	Unique project identifier.
milestone_id (FK)	INT	Foreign key linking to the Milestones table.
title	VARCHAR(150)	E.g., 'E-commerce Checkout System', 'Titanic Data Analysis'.
description	TEXT	Detailed description of the project goal (AI-generated template).
user_completion_status	ENUM	'Recommended', 'In Progress', 'Completed'.
user_github_link	VARCHAR(255)	Link to the Learner's live project or repository.
is_portfolio_visible	BOOLEAN	Flag for inclusion in the public portfolio.

Chapter 4

Scheduling and Estimates

The SkillVault project will be executed using an Agile methodology, broken down into three distinct phases. The initial phase focuses on the Minimum Viable Product (MVP) to validate the core AI-to-Roadmap functionality. The estimated total development time is **32 weeks** (8 months), excluding initial planning/discovery and final deployment/stabilization.

4.1 Phase I: Core MVP and Frontend (12 Weeks)

This phase establishes the foundational technical architecture and the core Learner experience (Goal → Roadmap → Tracking).

Table 4.1: Phase I: Core MVP and Frontend

Milestone	Tasks	Duration (Weeks)
1. Infrastructure	Setup Three-Tier Architecture, SQL Database, React environment. Implement secure user authentication (FR-1.0).	2
2. AI MVP	Implement initial Rule-Based AI Engine using static O*NET data. Implement Goal Definition and initial skill assessment (FR-2.0, FR-3.0).	3
3. Roadmap & Tracking	Implement AI Roadmap Generation (FR-4.0). Design and build Learner Dashboard with Progress Visualization (FR-8.0). Implement basic tracking logic (FR-7.0).	4
4. Resource Linking	Integrate curated external resource linking and project assignment framework (FR-5.0, FR-6.0). Finalize responsive UI design (NFR-U-1).	3
Total Phase I Duration		12

4.2 Phase II: Portfolio and Recruiter Features (14 Weeks)

This phase transforms the learning platform into a professional talent marketplace by integrating the automatic portfolio and the Recruiter Dashboard.

Table 4.2: Phase II: Portfolio and Recruiter Features

Milestone	Tasks	Duration (Weeks)
5. Portfolio Builder	Implement Automated Portfolio Creation logic (FR-9.0) and Portfolio Publishing (FR-10.0). Implement Portfolio Privacy Control (FR-11.0).	4
6. Recruiter UI	Develop Recruiter Dashboard Access (FR-12.0) and its dedicated, secure front-end interface.	3
7. Search Logic	Implement robust Candidate Search (FR-13.0) based on skill and project completion data. Develop Progress History View (FR-14.0).	4
8. Notifications & Polish	Implement Notification System (FR-15.0). Complete final system-wide security audit (NFR-S).	3
Total Phase II Duration		14

4.3 Phase III: Scaling and AI Enhancement (6 Weeks)

This final development phase focuses on enhancing performance, scaling the infrastructure, and advancing the AI from rule-based to a data-driven machine learning approach.

Table 4.3: Phase III: Scaling and AI Enhancement

Milestone	Tasks	Duration (Weeks)
9. Performance Tuning	Optimize database queries and API response times. Stress test for concurrency (NFR-P-3). Implement full error logging (NFR-R-3).	2
10. ML Integration Prep	Develop data ingestion pipelines for large-scale user progress data. Begin training an initial Machine Learning model to refine roadmap suggestions.	2
11. Deployment Prep	Finalize deployment scripts, run comprehensive integration testing, and prepare for public beta launch.	2
Total Phase III Duration		6

4.3.1 Summary of Estimates

- **Total Development Time:** $12 + 14 + 6 = 32$ weeks (8 months).
- **Resource Estimate (FTE):** Minimum project team of 3 Full-Time Equivalent (FTE) personnel: 1 Lead Developer/Architect, 1 Frontend/UX Specialist, 1 Backend/AI Engineer.
- **High-Level Project Risk:** The primary risk lies in the complexity and accuracy of the Machine Learning integration (Phase III). The project mitigates this by focusing on a robust, rule-based MVP in Phase I.